

Pavan Kumar Nayakanti

AI Cloud Solutions Architect | Enterprise AI Platform & Agentic AI | SLO-Governed Observability | GenAI at BFSI Scale | AWS | Multi-Cloud

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EXPERIENCE

Sr. Specialist - Cloud Practice & SRE Lead

IRIS Software - India (Clients: ADP, Charles Schwab, BMO)

 Sep 2020 - Present  Hyderabad, India

- **Enterprise AI Platform & RAG Architecture (BMO):** Designed and deployed GenAI assistants on AWS and Azure for BMO, using retrieval-augmented generation (RAG) pipelines for risk summarisation and interactive Q&A - Anthropic and Azure OpenAI models with compliance guardrails, audit trails, and BCRG/LTS retrieval infrastructure.
- **SLO Governance & Observability (ADP):** Defined and governed SLI/SLO frameworks and error budgets across multi-cloud AI and application platforms; deployed Prometheus, Grafana, Dynatrace, and Splunk for full-stack observability - reducing MTTR by 40%+ via automated remediation (Ansible + Python).
- **Reference Architectures & Cloud Guardrails:** Architected CPMI (Cloud Platform Maturity Index) and DevOps-as-a-Service frameworks establishing reference architectures, security guardrails, and cloud governance standards across ADP, Charles Schwab, and BMO.
- **IaC & Cloud-Native Platform (BMO):** Led end-to-end AWS migration using Terraform and AWS CDK - ECS Fargate container platforms, Lambda functions, and HA/DR architecture for .NET workloads; built compliance-ready CDK-enabled deployment pipelines.
- **Architecture Community & Team Scaling:** Scaled ADP SRE organization from 4 to 35 engineers across India, Canada, and the US; built Charles Schwab's 18-member Platform Engineering team from scratch; established architectural standards, design reviews, and engineering governance.
- **High-Throughput Deployment Platform (modeas):** Delivered 200+ builds/deployments per hour, eliminating 10,000+ manual operational hours annually and improving deployment cadence by 600%; executive-level reliability strategy presentations to VP and C-suite at ADP, Schwab, and BMO.
- **Chaos Engineering & Self-Healing:** Implemented Chaos Engineering with Keptn for proactive fault injection; built self-healing remediation frameworks converting reactive operations to predictive reliability - sustaining 99.9% release reliability at Charles Schwab.

SUMMARY

20+ year cloud and platform engineering leader specialising in enterprise AI platform architecture, cloud-native infrastructure (AWS, Azure, GCP), and SLO-governed observability at FinTech scale. Designed and deployed GenAI assistants and RAG pipelines on AWS and Azure for Tier-1 banks (ADP, BMO, Charles Schwab, RBC), with deep expertise in IaC (Terraform, AWS CDK), Kubernetes container orchestration, and enterprise security patterns (OAuth2, IAM, SOX/GDPR). Track record architecting reference architectures and cloud guardrails across 250+ applications in regulated financial services environments - now applying that foundation to enterprise AI platform and agentic AI architecture.

KEY ACHIEVEMENTS

-  **SRE Scale-Up: ADP**
Scaled SRE org from 4 to 35 engineers; cut MTTR by 40%+ with Dynatrace + Ansible automated remediation across multi-cloud platforms.
-  **modeas Deployment Platform**
200+ builds/hour; eliminated 10,000+ manual ops hours annually; 600% deployment cadence improvement.
-  **BMO GenAI Architecture**
Deployed RAG-based risk summarisation and Q&A on AWS/Azure for Tier-1 bank; compliance guardrails and audit-ready retrieval pipelines.
-  **RBC: 98% Onboarding Reduction**
Robo Deployment Framework cut application onboarding from 5 weeks to 5 hours; migrated 250+ apps across 60+ teams to GitHub.

Automation Technology Specialist

IRIS Software - Toronto, Canada (Client: Royal Bank of Canada Insurance)

Jun 2018 - Sep 2020 Toronto, Canada

- **Enterprise Integration & API Governance:** Governed enterprise-wide DevOps and change management across 250+ applications at RBC; enforced OAuth2 token management, SOX/GDPR compliance, and audit-ready change transparency through structured CI/CD pipelines and API integration patterns.
- **Identity & Security Patterns:** Enabled SAST, DAST, and IAST scanning as core DevSecOps objectives - integrating SonarQube, JFrog Artifactory, and GitHub into shift-left security pipelines; enforced IAM policies, network segregation, and authentication governance across 250+ application workloads.
- **DORA Framework & Observability:** Inducted DORA metrics across RBC Wealth Management and Insurance via Azure DevOps; implemented Splunk-based log aggregation and alerting pipelines integrated with CI/CD - establishing Deployment Frequency, MTTR, Change Failure Rate, and Lead Time as primary engineering health metrics.
- **Self-Service Deployment Platform:** Designed Robo Deployment Framework - PowerShell-native CI/CD orchestration enabling self-service release pipelines for Windows services, databases (SSIS, Oracle, SQL Server), and enterprise application workloads; reduced application onboarding from 5 weeks to 5 hours (98% reduction).
- Migrated 250+ applications across 60+ teams from CA SCM/Harvest to GitHub Enterprise, enabling modern CI/CD, agile collaboration, and self-service provisioning at scale.

Automation Architect

IRIS Software - Toronto, Canada (Insurance & Capital Markets)

Feb 2017 - Jun 2018 Toronto, Canada

- Designed fault-tolerant, container-orchestrated deployment pipelines using Kubernetes and Helm, enabling zero-downtime rollouts across capital markets platforms.
- Led RPA practice delivery automating 80+ applications and 250+ components across heterogeneous stacks (Java, NodeJS, Kafka, Salesforce, Mainframe); pioneered Cognitive and Intelligent Process Automation (CPA/IPA) through the Robotics Market Street programme.
- Closed \$478K DevOps engagement SOW (12-member team, 18-month engagement); enabled \$180K Capital Markets automation acquisition through pilot leadership at RBC Wealth Management.

Technology Manager

TechMahindra Pvt. Ltd. - India (Client: Adobe)

Jul 2015 - Feb 2017 Noida, India

- Architected and delivered zero-downtime business application migration during Adobe's data centre relocation; led SAP HANA upgrades and end-to-end data migration (Informatica to SQL Server 2012).
- Led cross-functional global engineering teams on large-scale enterprise delivery, aligning infrastructure initiatives with business timelines and stakeholder expectations.

Sr. SDET - Production Engineering & Platform Reliability

Microsoft Corporation - India

Jul 2007 - Dec 2014 Hyderabad, India

- Held dual mandate as Sr. SDET and Production Engineering owner for Sales & Marketing platforms serving tens of thousands of users across US, EMEA, and APAC - achieving 99.99% sustained uptime over 7 years.
- Defined SLO/SLI targets and error budgets; built QA automation frameworks integrated into CI/CD pipelines; modernised platforms via cloud migration and self-healing architectures supporting 200% user growth.

CERTIFICATIONS

AWS Certified Solutions Architect - Associate

Amazon Web Services - 2026

Azure Solutions Architect Expert (AZ-305)

Microsoft - 2026

Professional Cloud Architect (GCP)

Google Cloud - 2026

AI-102: Azure AI Solution Design

Microsoft - 2025

Certified Kubernetes Application Developer (CKAD)

CNCF - 2025

Certified SRE (iSRE)

iSRE Institute

EDUCATION

Masters in Artificial Intelligence 2024

Woolf University

General Management Program (GMP)

2022

ISB Hyderabad

M.Sc in Computer Science 2002

Osmania University